APRV Bedside Guidance 1.0 (WS 18/03/2020)

Commencing APRV for the novice

Table 1 Indications and contraindications for APRV

Indications	Contraindications (none are absolute)
Patients who are 'recruitable' (i.e. are considered to have collapsed/atelectactic areas of a lung)	Profound cardiovascular instability (particularly if secondary to untreated hypovolaemia)
Patients with diffuse disease process (e.g. ARDS and multifocal pneumonia)	Recent pulmonary resection with staple lines or anastomosis (i.e. postoperative lobectomy and pneumonectomy)
Patients in whom prone positioning is contraindicated or those who have not responded to a trial of prone positioning	Severe bronchospasm
Patients at high risk of respiratory deterioration (e.g. traumatic chest injury, aspiration, inhalational injury and pancreatitis)	Pulmonary hypertension with right ventricular decompensation (although a trial of APRV may be considered with close echocardiographic monitoring)
Patients with sepsis and multiple organ failure requiring invasive ventilation	Bronchopleural fistula
Obese patients	Untreated pneumothorax
Patients in whom a trial of low tidal volume ventilation has failed	Restrictive lung disease

Initiating APRV.

(i) Set FIo_2 (fraction of inspired oxygen) to 1.0, but this can usually be reduced rapidly.
(ii) Set P-high to current plateau pressure. In reality, this is usually $\leq$30 cmH_2O), but increased pressures may be necessary in morbidly obese patients.
(iii) Set P-low to 0 cmH_2O.
(iv) Set T-high to 5 s (other values can be used [range: 3–8 s], but we start with this value for ease of recollection).
(v) Set T-low initially to 0.5 s (range: 0.3–0.8 s).
(vi) Check that T-low is short enough to terminate expiratory flow at approximately 75% of peak expiratory flow. This is done by examining the flow/time graphical display on the ventilator, as shown in Fig. 3(A). If the expiratory flow terminates at <75% of peak expiratory flow, as demonstrated in Fig. 3(B), shorten T-low by steps of 0.1 s until the flow terminates at 75%. Similarly, a flow trace indicating that expiratory flow terminates too early will need prolongation of T-low in steps of 0.1 s.
(vii) Stop neuromuscular blocking agents and gradually reduce sedative drugs to encourage spontaneous breathing. The use of shorter-acting sedative medications allows for easier titration.
(viii) Tolerate hypercapnia, aiming to maintain pH $\geq$7.25 as with conventional ventilation (with the exception of neurocritical care patients).

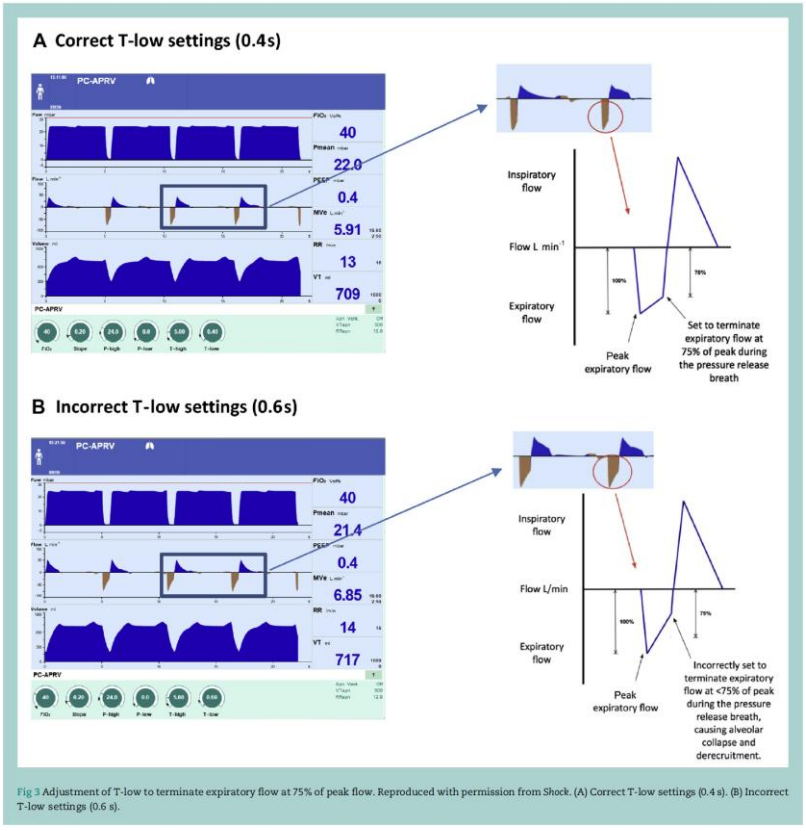


Fig 3 Adjustment of T-low to terminate expiratory flow at 75% of peak flow. Reproduced with permission from Shock. (A) Correct T-low settings (0.4 s). (B) Incorrect T-low settings (0.6 s).

NB in APRV tidal volumes frequently exceed 6ml/kg, and this is expected and not a concern

Review ABG at 4-6 hours, if not improved, abandon APRV, commence SIMV

Optimising APRV and Weaning

Table 2 Troubleshooting with APRV

Physiological derangement	Responses
Hyperoxaemia	(i) Reduce FIo_2 first. (ii) Once the FIo_2 is at 40–50%, start to reduce P-high.
Hypoxaemia	(i) Increase P-high by 2 cmH_2O. (ii) Increase T-high by 0.5–1 s. (iii) If T-high >10 s, consider reducing T-low by 0.2 s. (iv) Increase FIo_2.
Hypercapnia	(i) Tolerate hypercapnia if pH >7.25 and there are no adverse effects of acidosis. (ii) Ensure that the patient is making spontaneous ventilatory effort and aim to reduce sedation further to enhance this. (iii) Ensure that ATC (if applicable) is set to 100% with the correct tracheal tube size. (iv) Decrease T-high by 0.2 s down to a minimum of 3 s. (v) Check that ventilator circuit and heat and moisture exchanger filter are free of secretions or excessive moisture. (vi) Consider increasing P-high to maximise recruitment and minimise dead space.
Hypocapnia (assuming adequate cardiac output)	(i) Increase T-high by 0.2 s. (ii) If oxygenation is adequate or excessive, decrease P-high.
Hypotension	(i) Administer a fluid bolus of 250–500 ml crystalloid. (ii) Adjust or initiate vasoactive medicines. (iii) Consider urgent echocardiography to assess filling status and biventricular function.

Weaning APRV.

(i) Begin by reducing FIo_2.
(ii) Once FIo_2 has reduced to 0.4–0.5, start to reduce P-high.
(iii) Reduce P-high by 2 cmH_2O every 2–6 h whilst maintaining FIo_2 at 0.4–0.5. If this causes hypoxaemia, increase P-high by 4 cmH_2O and wean more slowly.
(iv) Once P-high is 20 cmH_20, increase T-high by 1–2 s each time P-high is decreased. Continuing this will wean the patient to CPAP <10 cmH_20 with minimal releases, allowing a treating clinician to assess a patient's readiness for extubation.
(v) Alternatively, once P-high is 12–15 cmH_2O with FIo_2 0.4, the patient can be switched to a conventional pressure supported mode, setting PEEP at the level of P-high with a low level of pressure support (e.g. 5 cmH_20) and weaned conventionally from there.

Guidance taken from Swindin, Sampson Howatson. 2020. Airway pressure release ventilation. BJA 20(3): 80-88.